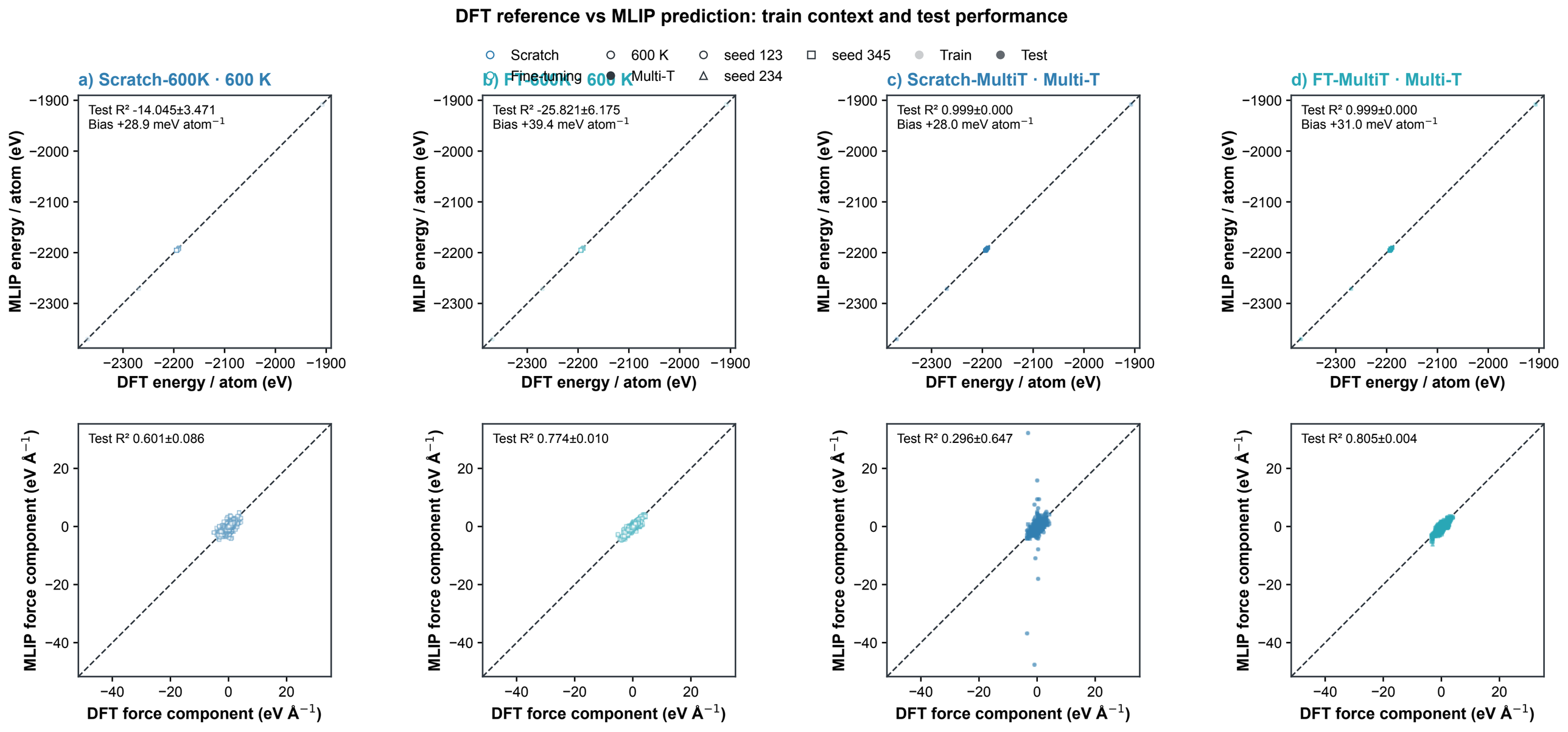

DFT reference vs MLIP prediction: train context and test performance
Scratch
Fine-tuning
600 K
Multi-T
seed 123
seed 234
seed 345
Train
Test
a) Scratch-600K · 600 K
b) FT-600K · 600 K
c) Scratch-MultiT · Multi-T
d) FT-MultiT · Multi-T
Test R² -14.045±3.471
Bias +28.9 meV atom⁻¹
Test R² -25.821±6.175
Bias +39.4 meV atom⁻¹
Test R² 0.999±0.000
Bias +28.0 meV atom⁻¹
Test R² 0.999±0.000
Bias +31.0 meV atom⁻¹
MLIP energy / atom (eV)
DFT energy / atom (eV)
Test R² 0.601±0.086
Test R² 0.774±0.010
Test R² 0.296±0.647
Test R² 0.805±0.004
MLIP force component (eV Å⁻¹)
DFT force component (eV Å⁻¹)